

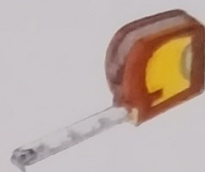
MEASURING LENGTH

Length is the measure of the distance between two points. A ruler is used to measure short lengths, while a metre rod or a metre tape is used for longer lengths. The distance between two cities is calculated using maps photographed by satellites.

A dressmaker uses a flexible metre tape to take measurements on a person. The salesperson uses a metre rod in a cloth store to measure the cloth. Civil engineers, carpenters, and surveyors use metre tapes made of steel, which are kept coiled inside a box.



metre tape



measuring tape
(made up of steel)



metre rod

Units of Measuring Length

In Class 3, we learnt about the basic unit of length—the metre (m) and two more units of length—the kilometre (km) and the centimetre (cm). Another unit of length is the millimetre (mm).

The thickness of a carrom striker, the width of a notebook, and so on, are measured in millimetres. The length of a notebook, or a pencil, and so on, are measured in centimetres. Short distances and lengths such as height of a tree or a building, length of cloth, and so on, are measured in metres. Kilometres are used to measure long distances. The distance between two cities is measured in kilometres.

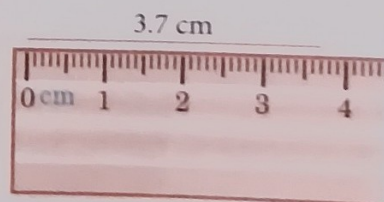
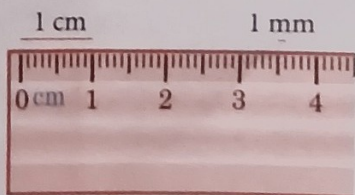


Line segments are measured using a ruler. The divisions on a ruler show centimetres and millimetres.

Note that each centimetre is divided into 10 small divisions. Each small division is a millimetre.

To measure a line segment, place the ruler such that the 0 mark is at one end of the line segment and the straight edge is along the line segment as shown here.

Read the marking at the other end of the line segment. If it is at 7 small divisions after 3 cm, then the line segment measures 3 cm 7 mm or 3.7 cm.



Let's Try! 1

Tick (✓) the units you would use to measure the following.

- Length of your foot
a. cm b. m c. km
- Distance from your front door to the kitchen
a. cm b. m c. km

Conversion of Units of Length

Measurement given in one unit can be converted to another by using the table given. The table shows how the units are compared with each other.

$1 \text{ m} = 1000 \text{ mm}$	$1 \text{ mm} = \frac{1}{1000} \text{ m} = 0.001 \text{ m}$	$1 \text{ km} = 1000 \text{ m}$	$1 \text{ m} = \frac{1}{1000} \text{ km} = 0.001 \text{ km}$
$1 \text{ m} = 100 \text{ cm}$	$1 \text{ cm} = \frac{1}{100} \text{ m} = 0.01 \text{ m}$	$1 \text{ cm} = 10 \text{ mm}$	$1 \text{ mm} = \frac{1}{10} \text{ cm} = 0.1 \text{ cm}$

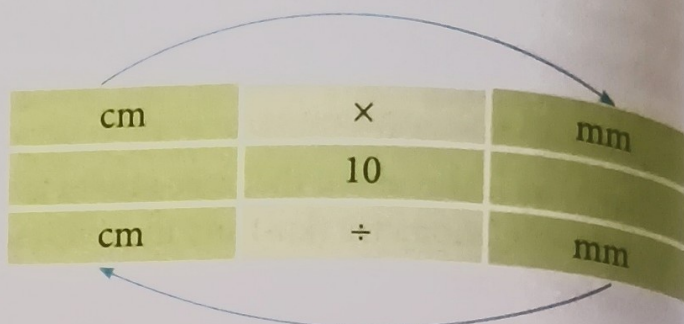
Centimetre to Millimetre and Millimetre to Centimetre

We know that,

$$1 \text{ cm} = 10 \text{ mm} \text{ and } 1 \text{ mm} = \frac{1}{10} \text{ cm}.$$

So, to convert centimetre to millimetre, we multiply by 10 and to convert millimetre to centimetre, we divide by 10.

Let us look at some examples.



Convert to millimetres	Convert to centimetres
$6 \text{ cm} = 6 \times 10 \text{ mm} = 60 \text{ mm}$	$38 \text{ mm} = \frac{38}{10} \text{ cm} = 3.8 \text{ cm}$
$7.9 \text{ cm} = 7 \text{ cm} + 9 \text{ mm}$ $= 70 \text{ mm} + 9 \text{ mm}$ $= 79 \text{ mm}$	$8 \text{ mm} = \frac{8}{10} \text{ cm}$ $= 0.8 \text{ cm}$
$38.6 \text{ cm} = 38 \text{ cm} + 6 \text{ mm}$ $= 380 \text{ mm} + 6 \text{ mm}$ $= 386 \text{ mm}$	$5980 \text{ mm} = \frac{5980}{10} \text{ cm}$ $= 598.0 \text{ cm}$ $= 598 \text{ cm}$

Let's Try! 2

Fill in the blanks.

- $45 \text{ cm} = \underline{\hspace{2cm}} \text{ mm}$
- $70 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$
- $999 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$

Key Point

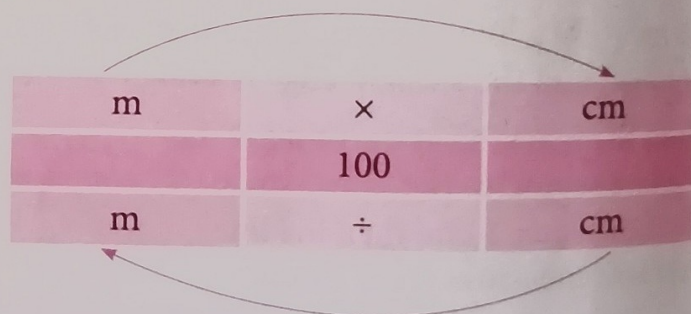
To convert millimetre to centimetre, we have to place the decimal point just to the left-hand side of the ones digit.

Metre to Centimetre and Centimetre to Metre

We know that,

$$1 \text{ m} = 100 \text{ cm} \text{ and } 1 \text{ cm} = \frac{1}{100} \text{ m}.$$

Hence, to convert metre to centimetre, we multiply by 100 and to convert centimetre to metre, we divide by 100.



Let us look at some examples.

Convert to centimetres	Convert to metres
$4 \text{ m} = 4 \times 100 \text{ cm} = 400 \text{ cm}$	$340 \text{ cm} = \frac{340}{100} \text{ m} = 3.40 \text{ m} = 3.4 \text{ m}$
$8.34 \text{ m} = 8 \text{ m } 34 \text{ cm}$ $= 800 \text{ cm} + 34 \text{ cm} = 834 \text{ cm}$	$88 \text{ cm} = \frac{88}{100} \text{ m} = 0.88 \text{ m}$
$11.09 \text{ m} = 11 \text{ m } 9 \text{ cm}$ $= 1100 \text{ cm} + 9 \text{ cm} = 1109 \text{ cm}$	$9 \text{ cm} = \frac{9}{100} \text{ m} = 0.09 \text{ m}$

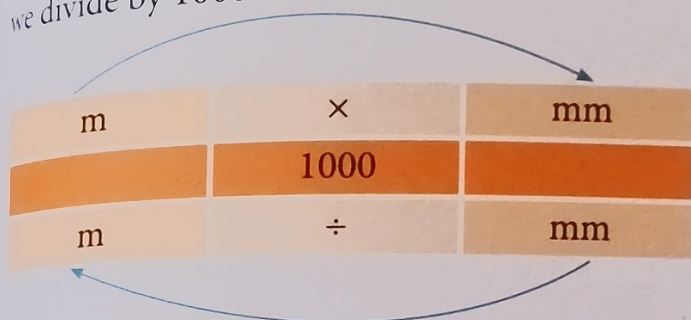
Key Point

When we convert centimetre to metre, the decimal point is placed just to the left-hand side of the tens digit. When there is no digit in the tens place, we write a 0 in the tens place and then place the decimal point.

Metre to Millimetre and Millimetre to Metre

We know that, $1 \text{ m} = 1000 \text{ mm}$ and $1 \text{ mm} = \frac{1}{1000} \text{ m}$.

To convert metre to millimetre, we multiply by 1000 and to convert millimetre to metre, we divide by 1000.



Let's Try! 3

Convert.

- 9 m into mm
- 1200 mm into m
- 3.9 m into cm
- 784 cm into m

Let us look at some examples.

Convert to millimetres	Convert to metres
$7 \text{ m} = 7 \times 1000 \text{ mm} = 7000 \text{ mm}$	$6000 \text{ mm} = \frac{6000}{1000} = 6 \text{ m}$
$20 \text{ m} = 20 \times 1000 \text{ mm} = 20000 \text{ mm}$	$7910 \text{ mm} = \frac{7910}{1000} \text{ m} = 7.910 \text{ m} = 7.91 \text{ m}$
$5.458 \text{ m} = 5 \text{ m } 458 \text{ mm}$ $= 5 \times 1000 \text{ mm} + 458 \text{ mm} = 5458 \text{ mm}$	$95 \text{ mm} = \frac{95}{1000} \text{ m} = 0.095 \text{ m}$

Key Point

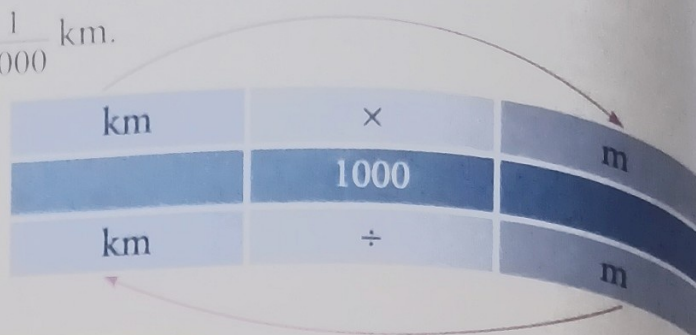
When we convert millimetre to metre, the decimal point is placed just to the left-hand side of the hundreds digit. When there is no digit in the hundreds place, we write a 0 in the hundreds place and then place the decimal point.

Metre to Kilometre and Kilometre to Metre

We know that, $1 \text{ km} = 1000 \text{ m}$ and $1 \text{ m} = \frac{1}{1000} \text{ km}$.

To convert kilometre to metre, we multiply by 1000 and to convert metre to kilometre, we divide by 1000.

Let us look at some examples.



Convert to metres	Convert to kilometres
$58 \text{ km} = 58 \times 1000 \text{ m}$ $= 58000 \text{ m}$	$9000 \text{ m} = \frac{9000}{1000} \text{ km} = 9 \text{ km}$
$12.75 \text{ km} = 12 \text{ km } 750 \text{ m}$ $= 12 \times 1000 \text{ m} + 750 \text{ m}$ $= 12750 \text{ m}$	$7865 \text{ m} = \frac{7865}{1000} \text{ km} = 7.865 \text{ km}$
$6.080 \text{ km} = 6 \text{ km } 80 \text{ m}$ $= 6000 \text{ m} + 80 \text{ m} = 6080 \text{ m}$	$75 \text{ m} = \frac{75}{1000} \text{ km} = 0.075 \text{ km}$

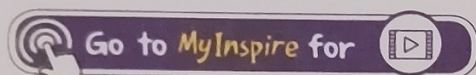
Let's Try! 4

Convert.

- 3892 m into km.
- 290 m into km.
- 39 km into m.
- 8 mm to m

Key Point

When we convert metre to kilometre, the decimal point is placed just to the left-hand side of the hundreds digit.



Exercise 8A

Go to MyInspire and attempt questions 1 to 4.

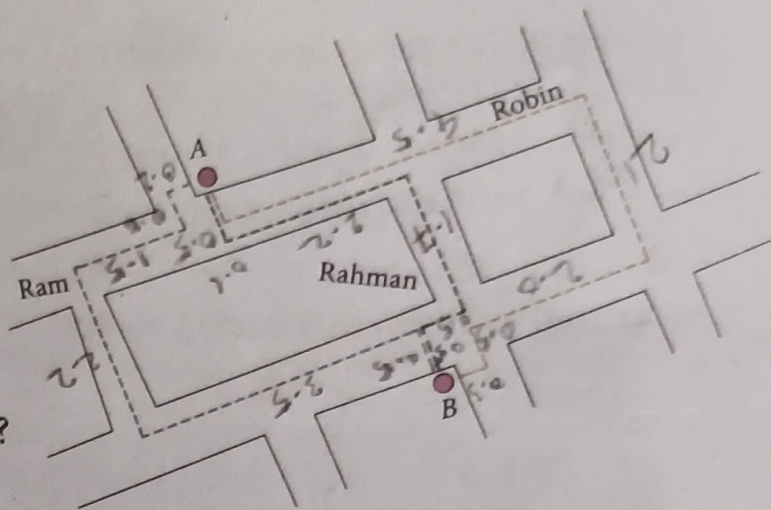
- Fill in the blanks with the correct unit—cm, m, or km.
 - The length of my pencil is about 7 cm.
 - The height of a table is about 2 m.

Application Based Questions

- Arrange in ascending order.
 - 45 mm; 67 mm; 3 cm; 89 mm; 4 cm 6 mm
 - 7 km; 6 km 99 m; 567 m; 2 km; 9 km 45 m
 - 123 m; 456 m; 6 km; 890 m; 3 km

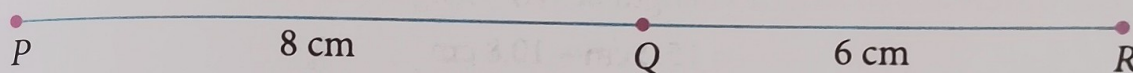
Ram, Robin, and Rahman walked from Point A to Point B, each one taking a different route. Use strings to measure the routes taken by them.

- Who walked the longest distance and who walked the shortest?
- What distance did each person walk if 1 cm on the string is equivalent to 50 m?



Addition and Subtraction of Length

Similar to numbers, lengths can be added or subtracted. Draw a line segment PQ measuring 8 cm and then extend it in the same direction to R so that QR measures 6 cm. The length of the line segment PR will be $PQ + QR = (8 + 6) \text{ cm} = 14 \text{ cm}$.



Let us look at some examples.

Example 1: Add the following.

- 35 m 17 cm and 8 m 25 cm

- 14 km 245 m and 23 km 176 m

Solution: Arrange vertically and add.

$$\begin{array}{r} \text{m} \quad \text{cm} \\ {}^1 35 \quad {}^1 17 \\ + 8 \quad 25 \\ \hline 43 \quad 42 \end{array}$$

$$\begin{array}{r} \text{km} \quad \text{m} \\ {}^1 14 \quad {}^1 245 \\ + 23 \quad 176 \\ \hline 37 \quad 421 \end{array}$$

Example 2: Subtract the following.

- 42 km 123 m from 78 km and 282 m
- 27 cm 5 m from 32 cm 61 m

Solution: Arrange vertically and subtract.

$$\begin{array}{r} \text{km} \quad \text{m} \\ 78 \quad 28^{12}2 \\ - 42 \quad 123 \\ \hline 36 \quad 159 \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ {}^2 32^{12}2 \quad {}^5 61^{11}1 \\ - 27 \quad 05 \\ \hline 5 \quad 56 \end{array}$$

Key Point

Since 1 m = 100 cm, we allot two digits for centimetre when we add or subtract.

Example 3: Mr Satish has to walk 80 m from his house to the bus stop. He takes a bus to his office. He travels 7 km 575 m in the bus to reach his office. What is the total distance travelled by Mr Satish to his office every day?

Solution: Total distance travelled = distance he walks + distance he travels by bus
 $= 80 \text{ m} + 7 \text{ km } 575 \text{ m}$

Arranging vertically and adding, we get

km	m
	1080
+ 7	575
7	655

Key Point

Since 1 km = 1000 m, we allot three digits for metre when we add or subtract.

So, Mr Satish travels 7 km 655 m every day to reach his office.

Example 4: Rakesh drew a line segment $WS = 15.2 \text{ cm}$ and Ramesh drew another line segment $PR = 10.8 \text{ cm}$. How much longer is WS than PR ? How much is this in millimetres?

Solution: The difference in length = length of WS - length of PR

$$= 15.2 \text{ cm} - 10.8 \text{ cm}$$

$$= 152 \text{ mm} - 108 \text{ mm}$$

$$= 44 \text{ mm}$$

	1	4	5	1	2
-	1	0	8		
		4	4		

So, WS is 44 mm longer than PR .

Let's Try! 5

1. Add 24 cm 5 mm and 15 cm 8 mm.

2. Subtract 5 m 18 cm from 9 m 5 cm.



Exercise 8B

1. Add the following.

a. $45 \text{ km } 977 \text{ m} + 6 \text{ km } 367 \text{ m}$

b. $36 \text{ m } 35 \text{ cm} + 16 \text{ m } 6 \text{ cm}$

c. $3 \text{ km } 675 \text{ m} + 2 \text{ km } 150 \text{ m} + 5 \text{ km } 5 \text{ m}$

d. $15 \text{ cm } 9 \text{ mm} + 45 \text{ cm } 5 \text{ mm} + 8 \text{ cm } 6 \text{ mm}$

2. Subtract the following.

a. $5 \text{ km } 645 \text{ m} - 3 \text{ km } 565 \text{ m}$

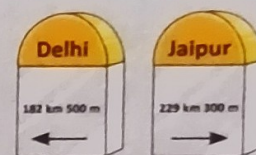
b. $15 \text{ m } 60 \text{ cm} - 9 \text{ m } 75 \text{ cm}$

c. $98 \text{ cm } 9 \text{ mm} - 22 \text{ cm } 8 \text{ mm}$

d. $497 \text{ m } 12 \text{ cm} - 289 \text{ m } 6 \text{ cm}$

Application Based Questions

- On a trekking expedition, Manohar climbed 2 km 875 m on the first day and 3 km 295 m on the second day. Find the total distance he climbed in the two days.
- Meena has to cycle 2 km 35 m to her school. In the evening she attends her music class which is situated at a distance of 780 m from the school. From there she has to cycle 1 km 415 m to reach home. Find the total distance she travels in a day.
- Murali Sreeshankar won the gold medal at the 12th International Jumping Meeting in Kallithea, Greece, with a jump of 8.31m. On the other hand, Thobias Montler won silver with an 8.27m jump. By what distance did Murali Sreeshankar win the long jump?
- Milestones are stones that indicate how far we have to go from a certain point to our destination. Two sides of a milestone are shown here. What is the distance between Delhi and Jaipur?



MEASURING MASS

In the metric system, the mass of the objects is measured in gram and kilogram. The basic unit of mass is the **gram**. Another units used to measure mass are **kilogram** and **Milligram**.

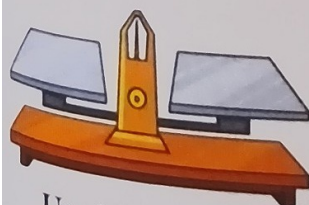
The mass of a very light or tiny object is measured in **milligrams**. For example, a gold ring, or a medicine capsule is measured in milligrams.

The mass of objects such as a box of chocolates, fruits, or vegetables is measured in **grams**.

The mass of objects such as a sack of vegetables or a packed suitcase is measured in **kilograms**.

We use **g** for gram, **kg** for kilogram, and **mg** for milligram.

Here are some pictures of machines that are used to weigh an object.



Used to weigh vegetables and fruits



Used in grocery stores, and supermarkets



Used to weigh very heavy objects

Let's Try! 6

Write the correct unit of measurement of the following.

- A sack of sugar
- A cup of sugar

Convert to grams

$$34 \text{ kg} = 34 \times 1000 \text{ g} = 34000 \text{ g}$$

$$7.45 \text{ kg} = 7.45 \text{ kg} = 7 \text{ kg } 450 \text{ g}$$

$$= 7 \times 1000 \text{ g} + 450 \text{ g}$$

$$= 7000 \text{ g} + 450 \text{ g} = 7450 \text{ g}$$

$$6.075 \text{ kg} = 6 \text{ kg } 75 \text{ g} = 6000 \text{ g} + 75 \text{ g}$$
$$= 6075 \text{ g}$$

Convert to kilograms

$$8000 \text{ g} = \frac{8000}{1000} \text{ kg} = 8.000 \text{ kg} = 8 \text{ kg}$$

$$38950 \text{ g} = \frac{38950}{1000} \text{ kg} = 38.950 \text{ kg}$$
$$= 38.95 \text{ kg}$$

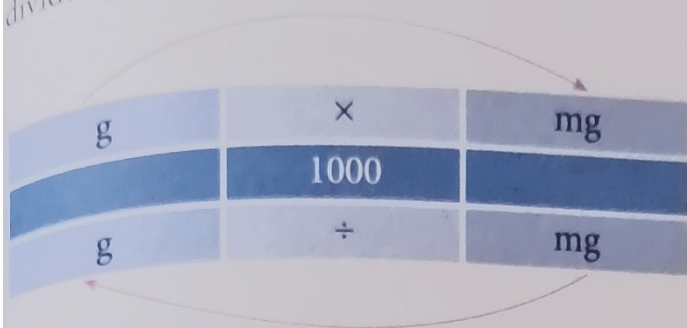
$$56 \text{ g} = \frac{56}{1000} \text{ kg} = 0.056 \text{ kg}$$

Key Point

When we convert gram to kilogram, the decimal point is placed just to the left-hand side of the hundreds digit.

Gram to Milligram and Milligram to Gram

To convert gram to milligram, we multiply by 1000 and to convert milligram to gram, we divide by 1000.



Let's Try! 8

Convert.

- 3892 mg into g
- 52.91 g into mg

Let us look at some examples.

Convert to milligrams	Convert to grams
$5 \text{ g} = 5 \times 1000 \text{ mg} = 5000 \text{ mg}$	$3000 \text{ mg} = \frac{3000}{1000} \text{ g} = 3 \text{ g}$
$4 \text{ g } 750 \text{ mg} = 4 \times 1000 \text{ mg} + 750 \text{ mg}$ $= 4000 \text{ mg} + 750 \text{ mg} = 4750 \text{ mg}$	$88000 \text{ mg} = \frac{88000}{1000} \text{ g} = 88 \text{ g}$
$16 \text{ g } 16 \text{ mg} = 16 \times 1000 \text{ mg} + 16 \text{ mg}$ $= 16000 \text{ mg} + 16 \text{ mg} = 16016 \text{ mg}$	$70 \text{ mg} = \frac{70}{1000} \text{ g} = 0.070 \text{ g} = 0.07 \text{ g}$

Key Point

When we convert milligram to gram, the decimal point is placed just to the left-hand side of the hundreds digit.

Example 5: A laptop weighs 2 kg. What is the mass in grams?

Solution: $1 \text{ kg} = 1000 \text{ g}$

Hence, $2 \text{ kg} = 2 \times 1000 \text{ g} = 2000 \text{ g}$.

The laptop weighs 2000 g.

Example 6: A bag of sugar weighs 25000 g. What is the mass in kilograms?

Solution: $1 \text{ g} = \frac{1}{1000} \text{ kg}$

Hence, $25000 \text{ g} = \frac{25000}{1000} \text{ kg} = 25 \text{ kg}$.

The bag of sugar weighs 25 kg.

Addition and Subtraction of Mass

As we did with lengths, we can also compare, add, and subtract masses. We should remember to express the masses in the same units.

Example 7: Gopal and his sister travelled to Mumbai by air to meet their parents. For domestic flights in India, one handbag up to 7 kg is allowed per customer for economy class. Gopal's handbag weighed 4 kg 150 g, while his sister Meena's handbag weighed 4455 g. Whose handbag was heavier and by how much?

Solution: Gopal's handbag weighs 4 kg 150 g which is the same as 4150 g.

Meena's handbag weighs 4455 g.

It can be seen that $4455 > 4150$.

$$4455 \text{ g} - 4150 \text{ g} = 305 \text{ g}$$

So, Meena's handbag was heavier by 305 g.

Example 8: The elevator in the Glowstar Mall can carry a maximum of 280 kg. There are four people already inside the elevator who weigh 75 kg, 65 kg, 40 kg, and 33 kg. Can a fifth person weighing 70 kg be allowed into the elevator?

Solution: The four people in the elevator weigh $75 \text{ kg} + 65 \text{ kg} + 40 \text{ kg} + 33 \text{ kg} = 213 \text{ kg}$

The elevator can carry a maximum of 280 kg.

Therefore, the fifth person can weigh $(280 - 213) \text{ kg} = 67 \text{ kg}$ or less than 67 kg

A person weighing 70 kg cannot be allowed into the elevator.

Example 9: Prema wishes to buy a gold ring weighing 2 g 50 mg and a necklace weighing 12 g 675 mg. If she has a 8 g gold coin which she wants to exchange for the jewellery, for how much gold does she have to pay?

Solution:

The jewellery weighs $2 \text{ g } 050 \text{ mg} + 12 \text{ g } 675 \text{ mg}$.

$$14 \text{ g } 725 \text{ mg} - 8 \text{ g of gold}$$

$$= 6 \text{ g } 725 \text{ mg of gold}$$

She needs to pay for 6 g 725 mg of gold.

Let's Try! 9

Solve.

$$\begin{array}{r} 1. \quad \text{g} \quad \text{mg} \\ 78 \quad 841 \\ + 52 \quad 621 \\ \hline \end{array}$$

$$\begin{array}{r} 2. \quad \text{kg} \quad \text{g} \\ 66 \quad 962 \\ - 31 \quad 562 \\ \hline \end{array}$$

Key Point

As $1 \text{ g} = 1000 \text{ mg}$, we have three digits for mg and we add or subtract as we do for numbers.



Exercise

4. Complete

a.

5. Convert

a. 5 kg

d. 10 kg

6. Convert

a. 500

d. 348

7. Solve the

a. 12 kg

c. 65 kg

e. 25 kg

f. 54 kg

Application

8. To prepare

20 kg 500 g

250 g of

9. The container

Textbook

file—1

weighs

10. In the

4 kg 500 g

of rice

the mo

Complete the following tables.

a.

kg	g
7	7000
12	12000
22	22000
3.9	3900

b.

g	mg
18	18000
2	2000
4	4000
0.5	500

Convert the following to grams.

- a. 5 kg 750 g b. 4 kg 10 g c. 8 kg 355 g
d. 10 kg 110 g e. 9 kg 950 g f. 6 kg 50 g

Convert the following to grams.

- a. 5000 mg b. 83000 mg c. 71000 mg
d. 34860 mg e. 65 mg f. 650 mg

Solve the following.

- a. 12 kg 350 g + 4 kg 65 g b. 56 g 450 mg + 12 g 980 mg + 7 g 225 mg
c. 65 kg 480 g - 39 kg 685 g d. 86 g 47 mg - 59 g 460 mg
e. 25 kg 460 g + 75 kg 10 g + 40 kg 750 g
f. 54 g 150 mg - 48 g 900 mg

Application Based Questions

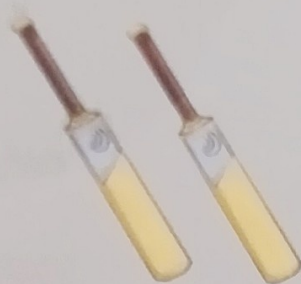
To prepare fruit salad for a wedding party, NG Caterers bought the following fruits: 20 kg 500 g of apples, 18 kg of pineapples, 22 kg 750 g of oranges, and 10 kg 250 g of grapes. How much did the fruits weigh in all?

The contents in Ramu's school bag are given below.

Textbooks—675 g, notebooks—415 g, pencil box—125 g, lunchbox—205 g, file—115 g. If the empty bag weighs 315 g, calculate how much the school bag weighs with its contents.

9. In the beginning of a month, Mrs Sharada had a stock of 25 kg of rice and 4 kg 500 g of dal. At the end of the month, she found that she was left with 8 kg 600 g of rice and 750 g of dal. Find how much rice and dal were consumed during the month.

11. Karthik had 2 cricket bats, 2 balls, a pair of gloves, a pair of shoes, and a pair of pads in his sports kit. Observe the pictures below and calculate the weight of his sports kit.



2 kg 450 g



150 g



80 g



770 g



415 g

12. Nimmi's sister weighed 3 kg 250 g at birth. After two months her weight was 4 kg 175 g. How much weight did she gain in two months?

MEASURING CAPACITY

Observe the following pictures.



Milk, oil, fruit juices, ice-cream, and honey are packed in containers such as bottles, tetra packs, or sachets. The volume of a container is the measure of how much it can hold. It is also known as the **capacity** of the container.

In the metric system, volume is measured in **litres** or **millilitres**.

The basic unit of volume is the **litre**. Similar to the conversion for the units of length and mass, 1 litre = 1000 millilitres.

We use L for litre and mL for millilitre.

Conversion of Units of Capacity

Now, let us learn how to convert litres to millilitres and vice versa.

Let's Try! 10

Fill in the blanks.

1. The volume of water in a cup is measured in _____.
2. The volume of water in a drum is measured in _____.

$$3 \text{ L } 850 \text{ mL} = 3000 \text{ mL} + 850 \text{ mL} = 3850 \text{ mL}$$

$$62000 \text{ mL} = \frac{62000}{1000} \text{ L} = 62 \text{ L}$$

$$12 \text{ L } 80 \text{ mL} = 12000 \text{ mL} + 80 \text{ mL} = 12080 \text{ mL}$$

$$76050 \text{ mL} = \frac{76050}{1000} \text{ L} = 76.050 \text{ L} = 76.05 \text{ L}$$

$$89 \text{ L } 5 \text{ mL} = 89000 \text{ mL} + 5 \text{ mL} = 89005 \text{ mL}$$

$$58 \text{ mL} = \frac{58}{1000} \text{ L} = 0.058 \text{ L}$$

Key Point

When we convert millilitre to litre, the decimal point is placed just to the left-hand side of the hundreds digit.

Example 10: How many millilitres is 6 L 300 mL?

$$\begin{aligned} \text{Solution: } 6 \text{ L } 300 \text{ mL} &= 6 \times 1000 \text{ mL} + 300 \text{ mL} \\ &= 6000 \text{ mL} + 300 \text{ mL} \\ &= 6300 \text{ mL} \end{aligned}$$

Example 11: How many litres is 8000 mL?

$$\text{Solution: } 8000 \text{ mL} = \frac{8000}{1000} \text{ L} = 8 \text{ L}$$

Let's Try! 11

- Convert from millilitre to litre.
a. 3231 mL b. 42 mL
- Convert from litre to millilitre.
a. 5.3 L b. 7 L

Addition and Subtraction of Capacity

Similar to lengths and masses, capacities that are in the same units can be added or subtracted.

Example 12: David adopted an orphan newborn puppy of weight 100 g. According to the advice of his veterinarian doctor, he gave a total of 91 mL of formula milk in the first week and 112 mL of formula milk in the second week. How much milk does he feed to his puppy in the two weeks?

$$\begin{aligned} \text{Solution: Total quantity of milk David feeds to his puppy in two weeks} &= 91 \text{ mL} + 112 \text{ mL} \\ &= 203 \text{ mL} \end{aligned}$$

Example 13: Rita's house has a rainwater harvesting system. During a monsoon, Rita filled a water tank with 1500 L of rainwater in the morning. 450 L 800 mL of water is used by the end of the day. What volume of water is left in the tank? Write at least two benefits of harvesting rainwater.

Solution: Volume of water left in the tank = $1500 \text{ L} - 450 \text{ L } 800 \text{ mL}$
 $= 1049 \text{ L } 200 \text{ mL}$

L	mL
1 ⁴ 5 ⁹ 0 ⁹ 0 ¹⁰	000
-	450 800
1049	200

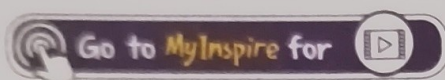
Key Point

As 1 L = 1000 mL, we allot three digits for mL and we add or subtract as we do for numbers.

Thus the volume of water left in the tank was 1049 L 200 mL.

Examples of the two benefits of harvesting rainwater are as follows:

- i. helps in maintaining the water level below the ground.
- ii. makes water available when there is less rainfall.



Exercise 8D

Go to MyInspire and attempt questions 1 to 3.

4. Complete the following table.

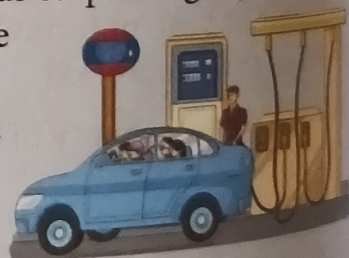
L	6	8	3	4	23	64
mL	6000	8000	3000	4300	23000	64000

5. Solve the following.

- | | |
|--|--|
| a. $32 \text{ L } 150 \text{ mL} + 8 \text{ L } 760 \text{ mL}$ | b. $45 \text{ L } 900 \text{ mL} + 6 \text{ L } 550 \text{ mL}$ |
| c. $58 \text{ L } 430 \text{ mL} - 39 \text{ L } 280 \text{ mL}$ | d. $97 \text{ L } 550 \text{ mL} - 75 \text{ L } 985 \text{ mL}$ |
| e. $7 \text{ L } 668 \text{ mL} + 14 \text{ L } 349 \text{ mL} + 23 \text{ L } 563 \text{ mL}$ | f. $39 \text{ L } 75 \text{ mL} - 28 \text{ L } 335 \text{ mL}$ |

Application Based Questions

6. Maya uses 2 L 560 mL of cooking oil per month. Meera uses 2650 mL and Mini uses 2 L 605 mL per month. Who uses the greatest and who uses the least amount of oil per month?
7. When two or more individuals share a car ride, it is known as carpooling. By carpooling, you can cut fuel and parking expenses; the more occupants you have, the more you save. Harish drives to his office every day. In March, he filled 12 L 675 mL of petrol in his car. In April, he opted to carpool with his office colleagues and filled 5 L 350 mL of petrol in his car.



- Find the total quantity of petrol he filled in his car in March and April.
- What other benefits do you think carpooling offers?

Mrs Jacob bought 2 L 500 mL of milk. She used 750 mL to make coffee and fed her cat with 350 mL of milk. Find the quantity of milk that is left over.



A barrel contains 45 L of diesel. It is poured into cans of capacity 4 L 500 mL, 10 L 650 mL, and 12 L 750 mL. Find the quantity of diesel still left in the barrel.

- In May and June, the temperature breaks all records in India. Even just standing outside for a short while is difficult.

To make the weather a little bit more bearable for others, Ram and his family decided to serve *sharbat* to passers-by on the roadside for a month. *Sharbat* is a sweet and energising cool drink. Each day they prepared *sharbat* with the following ingredients: rose syrup—10 L 250 mL, cooled water—800 L 780 mL, and milk—500 L 560 mL.



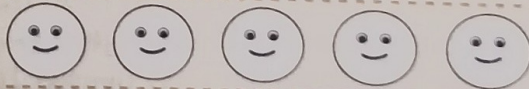
- Which of the quantities—cooled water or milk, used more and by how much to prepare sharbat every day?
 - Find the total quantity of rose syrup and milk was used to prepare the sharbat.
 - What moral value do Ram and his family display in the above scenario?
- Pratap had to take 5 mL of cough syrup once in the morning, afternoon, and at night. How much syrup does he take in a day? If he continued the same dosage of cough syrup for five days, how much medicine will be left in the bottle, if it contained 200 mL in the beginning?

Let's Review



Come here after you have completed the chapter. Colour 1 to 5 😊 to show how well you understand a topic. (5 😊 = Very well; 1 😊 = Not at all)

Measuring Length



Measuring Mass



Measuring Capacity



Revision Exercises



Start Up



1. Fill in the blanks.

a. $2 \text{ km } 130 \text{ m} = \underline{\hspace{2cm}} \text{ m}$

b. $534 \text{ cm} = \underline{\hspace{2cm}} \text{ m } \underline{\hspace{2cm}} \text{ cm}$

c. $5045 \text{ m} = \underline{\hspace{2cm}} \text{ km } \underline{\hspace{2cm}} \text{ m}$

d. $56 \text{ cm} = \underline{\hspace{2cm}} \text{ cm } \underline{\hspace{2cm}} \text{ mm}$

e. $5 \text{ kg } 132 \text{ g} = \underline{\hspace{2cm}} \text{ g}$

f. $34521 \text{ mL} = \underline{\hspace{2cm}} \text{ L } \underline{\hspace{2cm}} \text{ mL}$

g. $3201 \text{ g} = \underline{\hspace{2cm}} \text{ kg } \underline{\hspace{2cm}} \text{ g}$

h. $4 \text{ L} = \underline{\hspace{2cm}} \text{ mL}$

2. Convert the following.

a. 87 m into cm

b. $6 \text{ m } 15 \text{ cm}$ into cm

c. 234 mm into cm

d. 88 km into m

e. 600 cm into m

f. $12 \text{ km } 35 \text{ m}$ into m

g. 15 kg into g

h. $40 \text{ kg } 440 \text{ g}$ into g

i. 51050 mg into g

j. 11050 g into kg

k. $65 \text{ g } 10 \text{ mg}$ into mg

l. 65 mL into L

3. Solve the following.

a. $9 \text{ km } 200 \text{ m} + 3 \text{ km } 500 \text{ m}$

b. $9 \text{ m } 15 \text{ cm} + 6 \text{ m } 70 \text{ cm} + 6 \text{ m } 25 \text{ cm}$

c. $6 \text{ kg } 870 \text{ g} - 2 \text{ kg } 745 \text{ g}$

d. $25 \text{ L } 150 \text{ mL} - 12 \text{ L } 578 \text{ mL}$

Level Up



1. Arrange in ascending order.

a. 89 cm ; $4 \text{ m } 6 \text{ cm}$; $8 \text{ m } 8 \text{ cm}$; 109 cm ; 203 cm

b. 12 g ; 100 mg ; 200 mg ; 671 mg ; 10 g

2. Solve the following word problems.

a. Gopal Sharma undertook a pilgrimage on foot. He travelled $3 \text{ km } 550 \text{ m}$ on the first day, 5 km on the second day and $2 \text{ km } 675 \text{ m}$ on the third day. Find the total distance he walked on the three days.

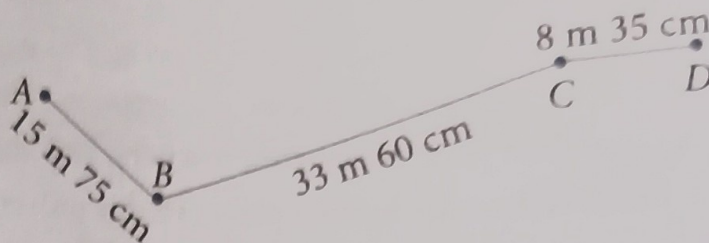
b. To complete a half-marathon race, a person would be expected to run a distance of 21 km . Avinash has covered $16 \text{ km } 245 \text{ m}$. Find the distance he has to run to complete the half-marathon race.

c. Sheetal's pet had a litter of 4 puppies weighing 165 g , 180 g , 145 g , and 154 g . What is the total weight of the pups? What is the difference in weight between the heaviest and the lightest of the pups?



A swimming pool can hold 62500 L of water. After cleaning the pool, water was being filled into the pool. When the volume of water pumped in was 46535 L, the pump broke down. How much more water was needed to fill the pool?

In the following image if you move from the point A to the point D, going through the points B and C on the way, what would be the total distance travelled by you?



OTS Questions



Starting from the top left corner, connect all the J's horizontally or vertically. You should not cut across a line that you have already drawn. You should reach the J at the bottom right corner. The path you take should be a continuous one. Assume that the distance you travel from one J to the next J is 25 cm. Calculate the total distance you have travelled to solve the puzzle! Give your answer in metres.

J	J	J	J	M	I	A
J	J	K	J	O	P	B
Y	S	T	J	J	J	Q
J	J	J	J	J	J	C
J	W	F	L	Z	N	H
J	J	J	J	J	J	J
X	B	G	D	J	J	J

Each of the athletes taking part in the school sports day events had 3 teaspoons of glucose mixed in 200 mL of water. One teaspoon of glucose weighs 4 g. If there were 40 athletes in all, find how many 2-L bottles of mineral water were used. Also, find the amount of glucose (in grams) that was mixed with the water for 40 athletes.

Cross-curricular Connect



(EVS) In EVS, you have learnt about the importance of potable water. The scarcity of clean potable water is a major environmental issue. You should help to conserve water by minimizing its usage and avoiding wastage.

Make a list of various activities for which you need water. Also, note down how much water is used for these activities. Find out the total amount of water you use every day. Discuss in class how you can minimize the wastage of water.

Have you ever faced a shortage of water in your house? When? What did you do then?

A tanker distributed 3000 L and 2750 L of water to two houses. If the capacity of the tanker is 7000 L, find the quantity of water left in the tanker.



Measurements play an important role throughout everybody's life, for example, in measuring the size of a bucket, the weight of bags, checking the size of footwear etc., Think of some more instances where you use the concept of measurement and discuss the same with your classmates, teachers, and family.

1. Earth is the only known planet in this universe where life is possible only because of the availability of water and oxygen. We use water for everything, from growing food to sanitation and recreation. So, we should not waste clean water and save it for future generations.



20 L 325 mL



10 L

In order to save water, Amayra decides to take a bucket bath instead of a shower bath. If she uses a bucket full of water for a bath, how much water does she save in a day?

2. Through travel, we learn about culture, food, new sites, music, and how people live their day-to-day lives in different parts of the world.

David and Peter live in the United States of America. Their friends live in Uttarakhand, state of India. The state has been blessed with gorgeous natural beauty, so they decide to visit them. David's travel bag weighs 14 kg and is 50 cm long. Peter's travel bag weighs 4 kg 250 g more and is 150 mm longer than David's travel bag. Find the mass and length of Peter's travel bag.



3. Visit a footwear store with your parents and make a note of the different types of footwear available for kids. Observe how a salesperson checks the size of the footwear that would fit a person. Find out your shoe size and the length of your foot in centimetres.

Convert the length of your foot to millimetres and also compare it with that of your father or mother. Collect the information about the various sizes that your classmates wear. Make a table to display the information.